

Eye-hand coordination develops from active multimodal compression

Francisco M. López^{1,2}, Bertram E. Shi³, and Jochen Triesch¹

¹ *Frankfurt Institute for Advanced Studies, Frankfurt, Germany*

² *Xidian-FIAS International Joint Research Center, Frankfurt, Germany*

³ *Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong*
{lopez,triesch}@fias.uni-frankfurt.de, eebert@ust.hk

Abstract—During their first months of life, infants learn to coordinate their perceptions and actions across different modalities. For example, eye-hand coordination relies on combining visual and proprioceptive sensory inputs for controlling eye and hand movements. What drives the development and calibration of such coordination? Here, we put forward a multimodal hierarchical extension of the Active Efficient Coding framework to learn a simple form of eye-hand coordination. By learning to actively compress visual and proprioceptive inputs into a combined multimodal representation, our embodied infant model learns to make eye movements to track an object held in its hand. We find that the abstract multimodal representation improves the tracking accuracy, but only if it emerges after the establishment of the single-modality systems. This suggests the existence of a “less-is-more” effect for the development of coordinated multimodal sensorimotor behaviors.

Index Terms—Eye-hand coordination, multimodality, active perception, sensorimotor development, less-is-more

I. INTRODUCTION

Humans are born with poor sensorimotor skills. During the first months of life, an infant will learn to control their own body and acquire complex coordinated behaviors by interacting with their environment, despite having scarce supervision. Many of these interactions result in rich multisensory events, where the consequences of actions of different actuators can be perceived across different modalities and this may be a key driving force behind learning coordinated behaviors.

As way of example, eye-hand coordination is first observed in neonates who extend their arms closer to an object [1] or move their hands into view [2]. At around 4 months of age, infants learn to make visually-guided reaching motions [3]. While it is often interpreted as a progressive development of a coordinated behavior [4], recent accounts challenge this view by suggesting initially independent developments of visual and somatosensory systems that are eventually aligned [5]. However, it is unclear how this multimodal coordination emerges [6].

In this work, we explore the development of eye-hand coordination in an embodied artificial agent, the multi-modal infant model MIMo [7]. Our goal is to shed light onto the role

played by multimodality in the alignment of the movements of the different effectors. In particular, we center our attention on the problem of tracking a hand-held object with the use of vision and proprioception. We propose that the autonomous self-calibration of eye-hand coordination can result from an active compression of multimodal information. The article is organized as follows: first, we provide an overview of related works. Next, we present our model, a multimodal extension of the Active Efficient Coding (AEC) framework. Then we explain the experiments carried out and show the results achieved, with focus on the consequences of a delayed and gradual onset of multimodality. Finally, we discuss the implications for developmental psychology.

II. RELATED WORK

Several studies have exploited multimodal representations to learn coordinated behavior. These include using a multimodal variational autoencoder to learn to imitate observed trajectories [8] and the development of self-touch through the joint encoding of proprioception and haptics [9]. To the best of our knowledge, our work is the first work to propose that active compression of multimodal signals gives rise to the development of eye-hand coordination.

The main interest in eye-hand coordination from the field of developmental robotics arises from a need to achieve visually-guided reaching. This requires precise estimates about the relative positions of the hand and the target. For example, [10] use deep neural networks to predict arm-torso configurations and possible contacts with a target from visual and proprioceptive inputs in an embodied robot simulation platform. Similarly, [11] perform arm pose estimations with a random regression forest method that uses depth images, trained on simulations but with successful performance on real robots. See [12] for a review of recent works in robotic reaching and grasping.

During the development of eye-hand coordination, infants may spend large amounts of time looking at their own hands [2], a phenomenon known as *hand regard*. Some studies aim to reproduce this with computational models. [13] uses a neural network to train hand regard in a simulated environment. Subsequently, the model is able to distinguish between the agent’s hand and a different stimulus. [14] train an intrinsically-motivated embodied agent to track its

This research was supported by “The Adaptive Mind” funded by the HMWK, Germany and by the Hong Kong Research Grant Council (Grant 16214821). JT was supported by the Johanna Quandt foundation.

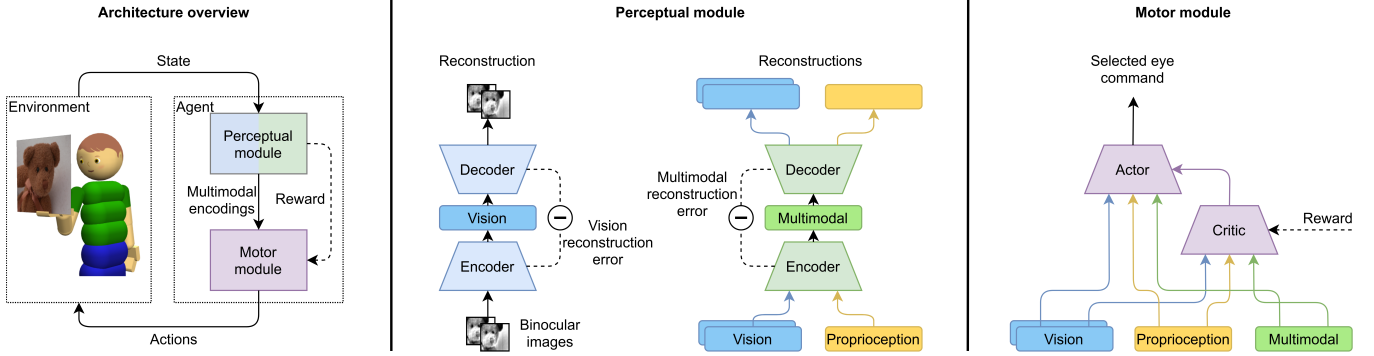


Fig. 1. Model architecture. *Left*: the environment consists of the infant embodiment MIMo [7] holding a target texture with his right hand. At every timestep, a state with visual and proprioceptive information is sent to the agent. This information is encoded by single-modality encoders and a multimodal encoder. Based on these learned encodings a motor module selects three eye movements: vergence, pan, and tilt. The agent computes its own intrinsic rewards based on the quality of the sensory encodings. *Center*: the perceptual module includes vision autoencoders (blue) where retinal images are compressed to create visual representations. A multimodal autoencoder (green) learns an abstract multimodal representation by jointly encoding vision, both the fine and coarse scales, and proprioception. *Right*: the motor module receives the multiple input modalities as states and the reconstruction errors as rewards.

own hand. The agent predicts the next state using vision and proprioception. Performance degrades when tracking an independently moving object, revealing the importance of proprioception.

III. MODEL

Our work is based on the theoretical framework of Active Efficient Coding (AEC), a generalization of the efficient coding hypothesis [15] that closes the perception-action loop with active perception. With AEC, sensorimotor systems can learn to self-calibrate by making actions towards states with high coding efficiency, i.e., states where the sensory signals can be compressed more easily. Using approximations to the mutual information between the sensory signals and their learned encoding as an intrinsic motivation for a reinforcement learning agent [16], AEC models can describe the autonomous self-calibration of active visual perception including vergence [17], smooth pursuit [18], and torsional eye movements [19].

Here, we propose the first hierarchical AEC multimodal model. Past work in multimodal AEC has integrated information either in parallel [20] or via prediction [14]. Wilmot et al. [21] show that the joint encoding and reconstruction of two or more sensory modalities preferentially retains the information that is shared across the different modalities, while the information that is unique to each modality is harder to recover. Following this idea, we propose that the joint encoding of multiple sensory inputs can lead to the autonomous self-calibration of multimodal coordination. In the case of eye-hand coordination, if the eyes are following the hand's movement, then the multimodal encoding of vision and proprioception should be more efficient than if the eyes are looking elsewhere. This is because visual and proprioceptive signals become partially redundant.

Following these principles, we develop a model that learns to visually track an object held by the hand through the use of vision, proprioception, and a more abstract and compressed multimodal encoding. Our model's architecture is shown in Fig. 1. We use an agent that at discrete timesteps samples states

from its environment and outputs eye movements as actions. The eye movements are of two types: vergence (rotation of the eyes in opposite directions to maintain binocular fixation); and smooth pursuit (horizontal and vertical rotations of the eyes in the same direction to track moving objects).

A. Environment

Our experiments are carried out using MIMo [7], a multimodal infant model based on the MuJoCo physics engine [22]. It consists of an embodiment of an 18-month old child with access to binocular vision, proprioception, haptics, and a vestibular system.

We place MIMo standing in an empty room holding a flat target texture in his right hand (see Fig. 1). This setting aims to recreate the experience of an infant that holds and looks at a toy. The target is a square of 10 cm that displays images from the *man-made* category of the *McGill Calibrated Color Image Database* [23]. MIMo moves the target at distances between 20 and 35 cm from MIMo's eyes, polar angles between 75° and 105°, and azimuth angles between 50° and 100°. These limits are chosen within the reachable range of MIMo's right hand. During training, MIMo moves the target randomly with polar and azimuth velocities of up to 1° per timestep, maintaining a constant radial distance. At every timestep, the target is rotated such that it is always facing MIMo's eyes. Our experiments seek to reproduce the motor babbling observed in infants and considered of key importance for the development of eye-hand coordination [1].

MIMo's visual system consists of two eyes separated by 4.9 cm. We simulate foveation of the eyes [24] by using two different scales: a fine scale with a field-of-view of 4°, and a coarse scale with a field-of-view of 8°. For each eye and scale, retinal images are rendered as 32×32 pixel grayscale images. Proprioception is represented as a vector of length 54, containing the angles of MIMo's joints relative to their resting state. The absolute position of the hand is not explicitly available, but it can be computed from the relative positions of the shoulder, elbow, and wrist joints. For added realism,

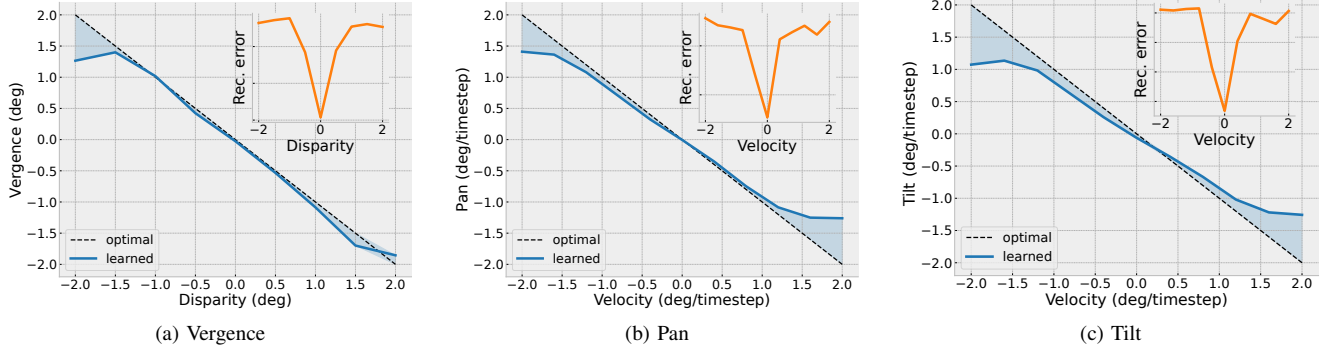


Fig. 2. Learned single-step policies. When testing each eye movement, the other two are manually set to follow their corresponding optimal policy. The blue shaded areas indicate the difference between the learned policy and the optimal policy, which is highest for the pursuit eye movements when the target moves at high relative velocities. The inset plots show the reconstruction errors of the joint autoencoders for the vergence pathway and pursuit pathway. In all cases, there is a minimum in the zero-disparity or zero-velocity states, allowing the agent to learn near-optimal policies.

we reduce the reliability of the proprioceptive input by adding zero-mean Gaussian noise with a standard deviation equal to 1/10 of the joint range. The visual and proprioceptive states are provided to the agent at every timestep; subsequently the selected eye movements are executed concurrently with the target's motion.

B. Perceptual module

The role of the perceptual module is twofold. First, it creates abstract representations of the sensory inputs that can be used by the motor module to select actions. Second, it computes the reconstruction errors of the representations. The reconstruction errors are used in cost functions to be minimized by both the perceptual and the motor modules.

A vergence pathway exploits binocular redundancies to guide vergence, and a pursuit pathway exploits temporal redundancies to guide pan and tilt eye movements [25]. For each of the two pathways, the perceptual module includes a vision autoencoder and a multimodal autoencoder (see Fig. 1).

The vergence vision autoencoder creates an abstract representation of the binocular images. The left and right images are concatenated on the color dimension to create a binocular image $\mathbf{i}$. It is passed to a convolutional encoder E^v with weights θ_E^v . Using the notation for convolutional layers of (number of filters, filter size, stride), the encoder consists of a first layer of (96, 8, 4) and a second layer of (24, 1, 1):

$$\mathbf{v} = E^v(\mathbf{i}; \theta_E^v). \quad (1)$$

The resulting encoding $\mathbf{v}$ is then passed to a transpose convolutional decoder D^v with weights θ_D^v consisting of a first layer of (96, 3, 2) and a second layer of (2, 4, 2):

$$\hat{\mathbf{i}} = D^v(\mathbf{v}; \theta_D^v). \quad (2)$$

Thus, a reconstruction $\hat{\mathbf{i}}$ of the original binocular image $\mathbf{i}$ is created, and a reconstruction error can be computed as the mean squared error between them:

$$\mathcal{L}^v = \text{MSE}(\mathbf{i}, \hat{\mathbf{i}}). \quad (3)$$

The vergence multimodal autoencoder creates a joint encoding from the vision encoding $\mathbf{v}$ and the proprioception input $\mathbf{p}$. The encoder E^m with weights θ_E^m has, for the vision encoding, a convolutional layer of (24, 3, 2) followed by a max pooling layer of filter size 2 and stride 2 and an additional fully connected layer of 1024 units; the proprioceptive input is passed through two fully connected layers of 128 units; then the resulting outputs are concatenated and passed through two fully connected layers of 1024 units:

$$\mathbf{m} = E^m(\mathbf{v}, \mathbf{p}; \theta_E^m). \quad (4)$$

The resulting multimodal encoding $\mathbf{m}$ enters the decoder D^m with weights θ_D^m . It has two fully connected layers with 1024 and 1152 units; the first 1024 values of its output are passed through a fully connected layer of 432 units, unflattened and upsampled with a factor 2, and passed through a transposed convolutional layer of (24, 2, 1) to reconstruct the vision encoding; the remaining 128 values are passed through two fully connected layers of 128 and 54 units to reconstruct the proprioception input:

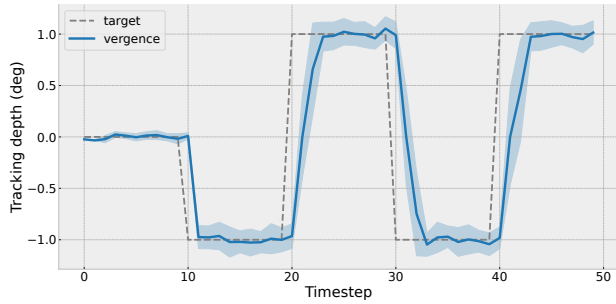
$$\hat{\mathbf{v}}, \hat{\mathbf{p}} = D^m(\mathbf{m}; \theta_D^m). \quad (5)$$

From the original visual and proprioceptive inputs $\mathbf{v}, \mathbf{p}$ and their reconstructions $\hat{\mathbf{v}}, \hat{\mathbf{p}}$, we compute a reconstruction error for the vergence multimodal autoencoder:

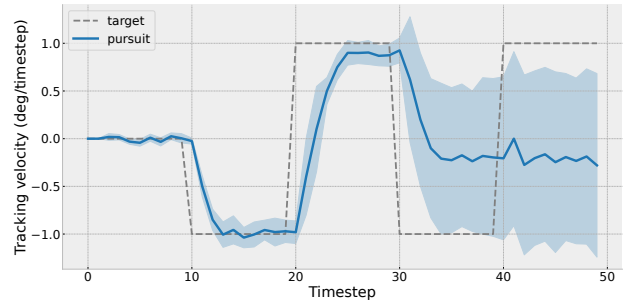
$$\mathcal{L}^m = \text{MSE}(\mathbf{v}, \hat{\mathbf{v}}) + \text{MSE}(\mathbf{p}, \hat{\mathbf{p}}). \quad (6)$$

The weights of the vision autoencoder, θ_E^v and θ_D^v , and of the multimodal autoencoder, θ_E^m and θ_D^m , are updated using the Adam algorithm [26], with a learning rate of 1×10^{-3} , using the corresponding reconstruction errors, $\mathcal{L}^v$ and $\mathcal{L}^m$ respectively, as loss functions.

The pursuit perceptual architecture is similar to the vergence architecture but with some key differences, since the pursuit pathway aims to exploit temporal redundancies. First, the input to the pursuit vision autoencoder are binocular images of two consecutive timesteps, resulting in four images stacked on the color dimension. Likewise, the inputs to the pursuit



(a) Depth tracking



(b) Motion tracking

Fig. 3. Adaptive behavior. *Left*: the target’s depth is changed every 10 timesteps, and the agent performs vergence actions to keep the image in focus. Throughout the duration of the testing episode, the agent is always able to find the target, even if it takes more than one timestep to do so. *Right*: the target’s diagonal velocity is changed every 10 timesteps while keeping a constant depth, and the agent performs pan and tilt actions to keep the image in focus. Initially the agent is capable of tracking the target, but after two velocity changes, roughly 30 timesteps, it is unable to do so as errors accumulate.

multimodal autoencoder are the pursuit vision encoding and two consecutive proprioceptive inputs. Furthermore, all the fully connected layers and number of convolutional filters are doubled with respect to the corresponding vergence ones.

The vision components of the perceptual module are doubled at the two visual scales, fine and coarse. The same networks are used for both cases, since the visual statistics in the environment are scale-invariant.

C. Motor module

The motor module uses the representations learned by the perceptual module at the fine and coarse scales to select actions. Following the principles of AEC, these actions aim to minimize the reconstruction errors of the autoencoders. More specifically, an agent that selects vergence actions at time t receives as a reward the weighted sum of the relative improvements of the vergence vision and multimodal reconstruction errors before and after making the eye movement:

$$r_t = c^v \frac{\mathcal{L}_{t-1}^v - \mathcal{L}_t^v}{\mathcal{L}_{t-1}^v} + c^m \frac{\mathcal{L}_{t-1}^m - \mathcal{L}_t^m}{\mathcal{L}_{t-1}^m}. \quad (7)$$

This agent has to learn a policy π to maximize the total return R_t over the duration of an entire episode:

$$R_t = \sum_{i=0}^{\infty} \gamma^i r_{t+i}, \quad (8)$$

where the discount factor γ ensures convergence. In our experiments, we use $\gamma = 0.6$, since no delayed rewards are possible during an episode. Alternative values lead to similar results with no significant differences. The reward function forces the agent to maximize two independent components, meaning this is a multi-objective learning problem. The values $c^v = 1$ and $c^m = 0.8$ were found to satisfactorily balance the two contributions across the state space.

The vergence agent is trained using the Soft Actor-Critic algorithm [27]. The actor and the critic are neural networks with weights θ_A^v and θ_C^v . Both receive the vision encoding $\mathbf{v}$, the proprioception input $\mathbf{p}$, and the multimodal encoding $\mathbf{m}$ as

an input. They have an initial convolution layer of (24, 2, 1) followed by a max pooling layer of filter size 2 and stride 2 and a fully connected layer of 1024 units for the vision encoding; two fully connected layers of size 128 for the proprioception input; and two fully connected layers of size 1024 for the multimodal encoding. All three outputs are then concatenated and passed through two fully connected layers of size 1024. The actor network outputs two values, a mean and a standard deviation, for a normal distribution from which the selected action is sampled. The critic network, that also receives the selected action as an input, outputs a single critic value, an estimate of the return, that is used to train the actor network.

Soft Actor-Critic favors guided exploration by balancing the maximization of the expected return R_t and entropy, i.e. the learned policy’s stochasticity, with a temperature parameter α . In line with [27] we use $\alpha = 0.2$. The network weights θ_A^v and θ_C^v are updated using the Adam algorithm with a learning rate of 3×10^{-4} .

Likewise, a smooth pursuit agent uses the pursuit inputs and selects two actions, the pan and tilt eye movements. It learns to maximize the total return for a reward equivalent to (7) computed with the pursuit reconstruction errors. All the parameters are the same as those of the vergence agent, but the network sizes are doubled.

IV. EXPERIMENTS

A. Multimodal compression gives rise to coordination

In a first experiment, we train the model for 100,000 epochs, with each epoch consisting of an episode of 10 timesteps. At the beginning of each episode, the target is placed at a random position and MIMO’s head is rotated to face the direction of the target. MIMO’s eyes are initialized with the pan and tilt angles set to zero, and a random vergence angle between 0 and 20°. Each vergence action consists of a change in the vergence angle between -2° and 2°; each pan and tilt actions change the pan and tilt velocities between -2°/timestep and 2°/timestep. The eyes are restricted to boundary values of 0° and 20° for the vergence angle, and -20° and 20° for the pan and tilt angles.

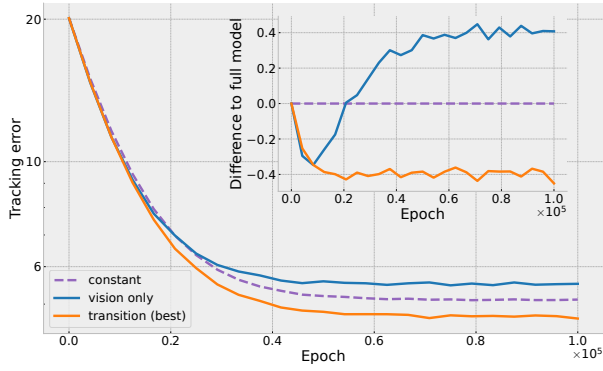


Fig. 4. Gradual development aids multimodal coordination. We compare the constant multimodal model shown in Fig. 1 with an ablated vision-only model (blue) and with a transition model from vision-only to multimodality, as detailed by (9). The vision-only model has a worse final performance but learns faster than the constant multimodal model. The transition model is able to outperform the constant model, thus highlighting a “less-is-more” effect. The inset shows the difference in performance relative to the constant model.

During each episode, the experiences are stored in a replay buffer of size 1,000 timesteps. At the end of the episode, all the network weights are updated using the procedures described previously, sampling a batch of size 200 from the buffer. Every 5,000 epochs we perform a performance test of the agent, where we evaluate the ability to track the target during a fixed trajectory and measure the cumulative absolute difference between the target center and the eyes’ fixation point.

After training, we perform a complete evaluation of the model. To do so, we use 20 new textures for the target, and place it in the default test position at a distance of 30 cm, polar angle of 90° , and azimuth angle of 90° . First, we evaluate the vergence reconstruction errors and vergence policy by manually setting the vergence angle between -2° and 2° relative to the target and measuring the model’s outputs. Second, we evaluate the pursuit reconstruction errors and pan and tilt policies by manually setting the eyes’ velocities between $-2^\circ/\text{timestep}$ and $2^\circ/\text{timestep}$ relative to the target’s velocity and measuring the model’s output. Importantly, when evaluating one eye movement, the other two are fixed to an optimal policy, so that their independent behaviors can be evaluated.

The policies learned by our model after training are shown in Fig. 2. For all three eye movements, the agent learns a near-optimal policy, thus reducing the angular or velocity disparity in a single timestep. This is a consequence of the reconstruction errors, which display sharp minima at the zero-disparity condition. According to (7), the reward for an action will be positive if the agent learns to move from a high reconstruction error state to a low one. The policies show greater inaccuracies for large disparities and velocities, which may be due to such states becoming increasingly rare during training, slowing down learning of the optimal action.

To further evaluate the model, we analyze its response to changes in the environment, as shown in the tracking curves of Fig. 3. When the distance of the target to MIMO changes, the agent can make vergence movements to track it, as predicted

by the policy function of Fig. 2. When the motion of the target changes, the agent can make pan and tilt movements that initially keep the target in the field of view. However, after two velocity changes, the agent fails to successfully keep track of the target. The reason for this failure may emerge from the duration of the evaluation, in comparison to the training episodes that only last 10 timesteps. After more than 20 timesteps, it is possible for the target to completely disappear from the field of view if it is not perfectly tracked, since errors can accumulate. In such cases, infants would make a saccade to find the target, which our model cannot make.

B. Less-is-more for multimodal sensorimotor learning

One of the goals of this work is to understand how the multimodal redundancies of vision and proprioception contribute to eye-hand coordination. To do so, we perform a second experiment exploring the idea of a development in stages. Initially each sensory modality develops on its own. Later they are combined [5].

We perform ablations of the model shown in Fig. 1, hereinafter referred to as *constant model*. First, we perform experiments with a vision-only alternative, where the motor module only receives the visual inputs and the vision reconstruction error as a reward. Second, we study a family of transition models that initially have vision-only inputs and reward, but where proprioception and the multimodal encodings and reward are gradually dialed in. As way of example, the reward in (7) is replaced by:

$$r_t = c^v \frac{\mathcal{L}_{t-1}^v - \mathcal{L}_t^v}{\mathcal{L}_{t-1}^v} + f_t c^m \frac{\mathcal{L}_{t-1}^m - \mathcal{L}_t^m}{\mathcal{L}_{t-1}^m}, \quad (9)$$

where the transition function f_t is given by:

$$f_t = \begin{cases} 0 & \text{if } t < \tau, \\ (t - \tau)/\Delta\tau & \text{if } \tau \leq t \leq \tau + \Delta\tau, \\ 1 & \text{if } \tau + \Delta\tau < t. \end{cases} \quad (10)$$

The agent transitions from vision-only for the first τ epochs to full multimodality after $\tau + \Delta\tau$ epochs. In our experiments we fix $\Delta\tau$ to 10,000 epochs and test different values of τ , the transition onset, between 0 and 100,000 epochs. In doing so, we can evaluate the need for an initial development of vision prior to the onset of multimodality.

In Fig. 4, we show that the model can learn to track the target using only vision, but performs worse than the constant model. We find that a transition from a vision-only model to a model with multimodal inputs and rewards outperforms the constant model, as long as the transition begins before 30,000 epochs (Fig. 5). Once the vision encodings are successfully learned, the multimodal encodings become useful to represent the state of the environment. Then the model can exploit multimodality to improve its coordination. An onset of 10,000 epochs provides the best performance.

V. DISCUSSION

This work describes the first computational model that exploits multimodal redundancies hierarchically to explain the

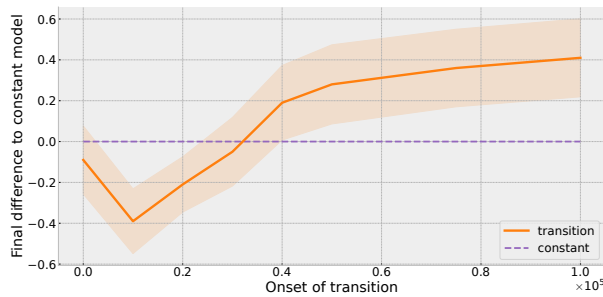


Fig. 5. Gradual development’s performance depends on transition onset. Given that the vision-only model learns faster than the constant model but has a worse final performance (see Fig. 4), a transition into multimodality can be beneficial if it starts during the early stages of training, i.e. up to 30,000 epochs. The best results are achieved with an onset of 10,000 epochs.

development of eye-hand coordination. In line with previous AEC models [17]–[19], [25], we find that it is possible to learn an autonomous self-calibration of the visual system from an intrinsic motivation to maximize the coding efficiency of visual inputs. We find that the additional motivation to maximize the efficiency of a multimodal joint encoding of vision and proprioception results in an enhancement of the coordinated behavior, but only if this multimodality emerges after adequate visual representations are learned. These results qualitatively confirm findings that the development of multimodal coordination requires prior development of single-modality systems [5]. This “less-is-more” approach to learning is well-understood in the context of language [28]. Our results suggest that multimodal sensorimotor development could benefit from staggered critical periods [29]. A delayed plasticity of higher-level, multimodal functions with respect to lower-level, single-modality processes may provide a computational advantage for the development of coordination.

We hope to extend this model to account for the main behavioral milestone of eye-hand coordination: reaching. Rather than spending too much time tracking their own hands [2], infants quickly move on to trying to touch and grasp new objects in their visual field [3]. We believe our work can provide the foundations for a model of the autonomous development of reaching. Likewise, the same principles can be applied to other examples of multimodal coordination.

REFERENCES

- [1] C. VonHofsten, “Eye–hand coordination in the newborn.” *Developmental psychology*, vol. 18, no. 3, p. 450, 1982.
- [2] A. L. van der Meer, “Keeping the arm in the limelight: Advanced visual control of arm movements in neonates,” *European Journal of Paediatric Neurology*, vol. 1, no. 4, pp. 103–108, 1997.
- [3] B. L. White, P. Castle, and R. Held, “Observations on the development of visually-directed reaching,” *Child development*, pp. 349–364, 1964.
- [4] J. Piaget, *The origins of intelligence in children*. International Universities Press New York, 1952, vol. 8, no. 5.
- [5] D. Corbetta, “Perception, action, and intrinsic motivation in infants’ motor-skill development,” *Current Directions in Psychological Science*, vol. 30, no. 5, pp. 418–424, 2021.
- [6] D. Corbetta, R. F. Wiener, S. L. Thurman, and E. McMahon, “The embodied origins of infant reaching: Implications for the emergence of eye-hand coordination,” *Kinesiology Review*, vol. 7, pp. 10–17, 2018.
- [7] D. Mattern, F. M. López, M. R. Ernst, A. Aubret, and J. Triesch, “Mimo: A multi-modal infant model for studying cognitive development in humans and ais,” in *2022 IEEE International Conference on Development and Learning (ICDL)*. IEEE, 2022, pp. 23–29.
- [8] M. Zambelli, A. Cully, and Y. Demiris, “Multimodal representation models for prediction and control from partial information,” *Robotics and Autonomous Systems*, vol. 123, p. 103312, 2020.
- [9] V. Marcel, J. K. O’Regan, and M. Hoffmann, “Learning to reach to own body from spontaneous self-touch using a generative model,” in *2022 IEEE International Conference on Development and Learning (ICDL)*. IEEE, 2022, pp. 328–335.
- [10] P. D. Nguyen, M. Hoffmann, U. Pattacini, and G. Metta, “Reaching development through visuo-proprioceptive-tactile integration on a humanoid robot—a deep learning approach,” in *Joint International Conference on Development and Learning and Epigenetic Robotics*, 2019, pp. 163–170.
- [11] F. Widmaier, D. Kappler, S. Schaal, and J. Bohg, “Robot arm pose estimation by pixel-wise regression of joint angles,” in *2016 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2016, pp. 616–623.
- [12] R. Newbury, M. Gu, L. Chumbley, A. Mousavian, C. Eppner, J. Leitner, J. Bohg, A. Morales, T. Asfour, D. Kragic *et al.*, “Deep learning approaches to grasp synthesis: A review,” *IEEE Transactions on Robotics*, 2023.
- [13] T. Homma, “Hand recognition obtained by simulation of hand regard,” *Frontiers in psychology*, vol. 9, p. 729, 2018.
- [14] L. P. Wijesinghe, J. Triesch, and B. E. Shi, “Robot end effector tracking using predictive multisensory integration,” *Frontiers in neurorobotics*, vol. 12, p. 66, 2018.
- [15] H. B. Barlow *et al.*, “Possible principles underlying the transformation of sensory messages,” *Sensory communication*, vol. 1, no. 01, pp. 217–233, 1961.
- [16] A. Aubret, L. Matignon, and S. Hassas, “A survey on intrinsic motivation in reinforcement learning,” *arXiv preprint arXiv:1908.06976*, 2019.
- [17] Y. Zhao, C. A. Rothkopf, J. Triesch, and B. E. Shi, “A unified model of the joint development of disparity selectivity and vergence control,” in *2012 IEEE International Conference on Development and Learning and Epigenetic Robotics (icdl)*. IEEE, 2012, pp. 1–6.
- [18] C. Teulière, S. Forestier, L. Lonini, C. Zhang, Y. Zhao, B. Shi, and J. Triesch, “Self-calibrating smooth pursuit through active efficient coding,” *Robotics and Autonomous Systems*, vol. 71, pp. 3–12, 2015.
- [19] Q. Zhu, C. Zhang, J. Triesch, and B. E. Shi, “Learning torsional eye movements through active efficient coding,” *Neuromorphic Computing and Engineering*, vol. 2, no. 3, p. 034007, 2022.
- [20] C. Zhang, J. Triesch, and B. E. Shi, “Learning multisensory cue integration on mobile robots,” in *IEEE International Conference on Robotics and Automation*. IEEE, 2017.
- [21] C. Wilmot, G. Baldassarre, and J. Triesch, “Learning abstract representations through lossy compression of multi-modal signals,” *IEEE Transactions on Cognitive and Developmental Systems*, 2021.
- [22] E. Todorov, T. Erez, and Y. Tassa, “Mujoco: A physics engine for model-based control,” in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2012, pp. 5026–5033.
- [23] A. Olmos and F. A. A. Kingdom, “McGill calibrated colour image database,” <http://tabby.vision.mcgill.ca>.
- [24] C. Weber and J. Triesch, “Implementations and implications of foveated vision,” *Recent Patents on Computer Science*, vol. 2, pp. 75–85, 2009.
- [25] C. Wilmot, B. E. Shi, and J. Triesch, “Self-calibrating active binocular vision via active efficient coding with deep autoencoders,” in *2020 Joint IEEE 10th International Conference on Development and Learning and Epigenetic Robotics (ICDL-EpiRob)*. IEEE, 2020, pp. 1–6.
- [26] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [27] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Intl. Conf. on Machine Learning*, 2018, pp. 1861–1870.
- [28] E. L. Newport, “Maturational constraints on language learning,” *Cognitive science*, vol. 14, no. 1, pp. 11–28, 1990.
- [29] R. K. Reh, B. G. Dias, C. A. Nelson III, D. Kaufer, J. F. Werker, B. Kolb, J. D. Levine, and T. K. Hensch, “Critical period regulation across multiple timescales,” *Proceedings of the National Academy of Sciences*, vol. 117, no. 38, pp. 23 242–23 251, 2020.